Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/09/2011
PI Name:	Freed, Lisa E M.D., Ph.D.		
Project Title:	Microgravity Tissue Engineering		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02139-4037	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2010
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 11/30/2010 per PI/JSC approval notice (09/2010) NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Received NCE to 11/30/2009 (from 11/30/2008) per PI (11/08) NOTE: Received NCE to 11/30/2008 (from 11/30/2007) per P. Denkins/JSC (7/07) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	01-25-2010: Co-Is Vunjak-Novakovic, Rupnick, and Padera were not active during 2009-2010.		
COI Name (Institution):	Vunjak-Novakovic, Gordana (Columbia University) Rupnick, Maria (Harvard Medical School) Guilak, Farshid (Duke University Medical Center) Padera, Robert (Harvard Medical School)		
Grant/Contract No.:	NNJ04HC72G		
Performance Goal No.:	1.21		

Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.
Task Description:	Our research involves cell- and tissue-levels studies of 3D engineered tissue constructs and utilizes models of relevance to tissue loss in astronauts in microgravity and skeletal and myocardial tissue regeneration in human patients on earth. We selected to study skeletal and cardiac tissue regeneration because: (a) spaceflight adversely affects these tissues, (b) improved treatments are needed for millions of patients who suffer from progressive skeletal and cardiac diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli on cells and tissues, and (d) we have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer. Specifically, we designed in vitro models based on cells, biomaterial scaffolds, and advanced bioreactor systems, and showed bioreactor-grown constructs were larger and functionally superior to controls grown using conventional culture dishes. The current limitations of our tissue engineering research are as follows: (i) long periods of in vitro culture are required to generate mechanically functional engineered cartilage, (ii) heart cell viability and contractility are poor in engineered heart muscle, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. We are addressing these limitations by: (a) integrated studies in cell and molecular biology, mass transport, design and operation of bioreactors including microgravity analog systems, tissue physiology and biomechanics, and implant biocompatibility and remodeling, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales. The project is organized as two Aims: (1) skeletal tissue engineering, and (2) cardiac tissue engineering. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We tested this hypothesis in controlled studies of tissue engineered constructs, i.e., cells cultured on 3D scaffolds, in the presence of biophysical stimuli. Critical system components include the cells (primary and stem cells), scaffolds (biodegradable biomaterials), and in vitro/in vivo models to assess construct properties and capacity for defect repair. Specifically, tissue engineered constructs are assessed with respect to cell viability, tissue structure, molecular composition, and mechanical and electrophysiological function. Construct properties are correlated with in vitro culture conditions and the outcome of in vivo studies to provide feedback necessary to tailor model system parameters and obtain the best overall result. The outcomes are improved understanding of how biophysical stimuli regulate skeletal and cardiac tissues, and the generation of mechanically functional tissue engineered constructs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our combined use of 3D engineered tissue constructs and advanced bioreactor systems provides a unique model system for controlled in vitro studies at the cell and tissue-levels that may be relevant to treating skeletal and myocardial tissue loss and dysfunction in astronauts. Biophysical stimuli play a critical role in tissue growth, maintenance, and remodeling. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations can cause fluid and electrolyte imbalance, altered neurohormonal regulation, diminished tissue remodeling, and loss of bone and muscle mass. Exercise regimens and pharmaceutical interventions have been proposed to try to maintain skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability to determine the underlying mechanism(s) or assess the potential risks/benefits of currently used countermeasures is limited by a lack of experimental models. Our controlled in vitro studies of how 3D tissue engineered constructs respond to biophysical stimuli are expected to help identify how the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding can reasonably be expected to aid in the development of better countermeasures. Our tissue engineering research has implications not only for astronauts but also for humans on earth, e.g., by developing in vitro analogs for experimental pharmaceutical studies, and new treatments for skeletal and cardiovascular diseases. Skeletal and cardiac tissues perform structural and mechanical functions that are vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited healing capacity, and no currently available therapy can predictably restore a functional articulating surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle, and heart disease is the leading cause of morbidity and mortality in developed countries. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting 3D tissue constructs that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, tissue engineered grafts for cartilage and cardiac regeneration still face many challenges in order to be suitable for widespread clinical use, hence this research can clearly benefit human patients on earth.
	Microgravity research and microgravity-related technologies have helped to identify how spaceflight compromises the ability of normal tissues to withstand mechanical forces, and fostered pioneering studies of cell biology and tissue engineering. We were very fortunate that NASA funded a series of four research projects, initially entitled “Microgravity studies of cell-polymer cartilage implants” and subsequently, “Microgravity tissue engineering.” During the first project period (1992-1995), an experimental model involving 3D tissue engineered constructs and tissue culture bioreactors was conceived and implemented. During the second project period (1995-1998), a three month long experiment “Tissue engineering of cartilage in space” was carried out on-board STS-79, the Mir Space Station and STS-81 (Freed et al., Proc. National Academy of Sciences, USA, 1997). Also, the experimental model was expanded to include bone tissue and heart muscle as well as cartilage. During the third (1999-2004) and current (2003-2010) project periods, 3D tissue engineered cartilage, bone, and heart muscle constructs were used for integrated studies of cell and molecular biology, tissue physiology, biomechanics, integration and remodeling. Of note, during NASA’s implementation of the “vision for space exploration” announced by President Bush in January 2004, funding of the current project period was discontinued in 2005 and ultimately restored in 2006. Nevertheless, the current project period involved three co-investigators, and contributed to the training of ten postdoctoral researchers and four graduate students, as well as undergraduate students. Moreover, the current project period yielded 28 publications in leading scientific journals, three book chapters, 41 conference presentations, and two patent applications. The research has helped define how biophysical cues regulate skeletal and cardiac tissue development and function, and yielded 3D tissue

Task Progress:	engineered skeletal and cardiac constructs with superior mechanical functionality than was previously reported in the literature. In addition to NASA rotating bioreactors operated at 1g, complementary bioreactors providing mechanical or electrical stimulation, direct perfusion, and oscillatory fluid flow were developed leading in one case to a licensed patent application. Skeletal Tissue Engineering Cartilage tissue engineering is currently limited by failure of cell-based constructs to mimic the mechanical and structural properties of native skeletal tissues (Freed et al., Advanced Materials, 2009). Toward addressing this challenge, we cultured cartilage cells and bone marrow-derived stem cells on biomaterial scaffolds in the presence of biophysical signals in vitro. In an attempt to recreate the complex mechanical properties of normal articular cartilage, a 3D weaving technique we previously developed and demonstrated (Moutos et al., Nature Materials, 2007). Specifically, 3D-woven poly(glycolic acid) yarns were infiltrated with a hydrogel, and the resulting constructs displayed aggregate and tensile moduli within one order of magnitude of normal cartilage, with significant anisotropy in failure stress, failure strain, tangent modulus, and energy-to-failure. During the present project period, we engineered cartilage (Valonen et al., Biomaterials, 2010) and bone-like (Abrahamsson et al., Tissue Engineering 2010) constructs were generated by culturing bone-marrow derived adult human mesenchymal stem cells on 3D woven poly(e-caprolactone) scaffolds. Significant independent and interactive effects of growth factors and an oscillating perfused bioreactor on tissue development were demonstrated over a period of three to six weeks. Mechanical testing of the resulting constructs showed that the 3D woven scaffold provided robust mechanical properties, suggesting that engineered cartilage based on 3D woven scaffolds has the potential to provide mechanical function immediately after cell seeding, instead of requiring prolonged culture in vitro. Our ongoing work involves osteochondral tissue engineering, starting from human mesenchymal stem cells, 3D woven scaffolds and a perfused bioreactor (Larson et al., unpublished work). Cardiac Tissue Engineering Cardiac tissue engineering is currently limited by inadequate survival and contractility of component heart cells, and failure of traditional 3D scaffolds to mimic the mechanical and structural properties of native cardiac tissue (Freed et al., Advanced Materials, 2009). Toward addressing these challenges, we cultured heart cells on biomaterial scaffolds in the presence of biophysical signals in vitro. We also used an oscillatory perfused bioreactor to demonstrate individual and interactive effects of growth factors and perfusion on engineered cardiac constructs based on heart cells and collagen sponges (Cheng et al., Tissue Engineering, 2009). Specifically, slow, bi-directional perfusion and IGF enhanced cell viability, improved construct electrophysiologic properties, and increased amounts of cardiac marker proteins and total protein. These findings demonstrate that application of multi-factorial stimuli to developing engineered cardiac constructs can affect the quality of the resulting tissue. Toward mimicking the structural and mechanical properties of normal heart muscle, we used laser microablation to produce elastomeric poly(glycerol-sebacate) (PGS) scaffolds with accordion-like honeycomb pores and directionally dependent structural and mechanical properties (Engelmayr et al., Nature Materials, 2008). By controlling polymer curing time and scaffold pore design, we closely matched tensile moduli and anisotropy ratio of the scaffolds to those of normal heart muscle and increased the orientation of cultured heart cells. In another study, we demonstrated feasibility of using PGS scaffolds for myocardial defect repair by implanting cell-free scaffolds on the surface of a rodent heart following surgically induced myocardial infarction (Radisic et al., J. Biomedical Materials Research, 2008). Ongoing work focuses on integrated experimentation and mathematical modeling toward rational design of cardiac tissue engineering scaffolds. New Technology We described an oscillatory perfused bioreactor that provided effective convective-diffusive mass transport of gases, nutrients, and regulatory molecules to developing tissue constructs (US Patent Application No. 12/526,300), and demonstrated utility of this device for two tissue engineering applications: heart muscle (Cheng et al., Tissue Engineering, 2009) and cartilage (Valonen et al., Biomaterials, 2010). [Editor's note: To find referenced citations prior to 2010, see bibliographies in PI's earlier reports.]
Bibliography Type:	Description: (Last Updated: 09/14/2021)
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Patents	US Patent Application 2010/0297233 A1 filed February, 2008. Licensed September, 2009 ; November 2010. Nov-2010 Moretti M, Freed LE, Langer R. "Oscillating Cell Culture Bioreactor."
Significant Media Coverage	FLL Global. "In 2009-2010, Lisa Freed served as a member of the FLL Global Challenge Advisory Team for the 2010 'Body Forward' competition in biomedical engineering at the FIRST LEGO League (FLL)." FLL posted an interview of Dr. Freed on You Tube (http://www.youtube.com/watch?v=XUhR8mOaEVk&feature=youtube_gdata), July 2010., Jul-2010